

Skills Summary

Congruent or Similar? Evidence from Transformations

Use this reference while completing your worksheet or when studying.

The two tools. Distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and slope $m = \frac{y_2 - y_1}{x_2 - x_1}$ (undefined if the run is 0).

What each proves. Equal corresponding *lengths* $\Rightarrow$ congruent (some rigid motion works). Proportional corresponding lengths, all with the same factor $k \Rightarrow$ similar. *Slopes* are what tell you **which** motion: translation keeps every slope; a half turn keeps every slope; a quarter turn gives the negative reciprocal; a reflection across $y = x$ gives the reciprocal.

1. Translation: slide, nothing else changes

Rule: $(x, y) \rightarrow (x + a, y + b)$, the same a and b for every point. Evidence to collect: corresponding lengths are equal *and* corresponding slopes are equal, so each side of the image is parallel to its partner. If the lengths match but a slope has changed, the motion was not a translation.

Example: $A(1, 2)$ and $B(5, 5)$ slide by $(x, y) \rightarrow (x + 3, y - 2)$, giving $A'(4, 0)$ and $B'(8, 3)$. Find $A'B'$.

Step 1. The claim is that lengths are unchanged, so compute the image length and compare it with $AB = 5$ — that comparison is the evidence.

Step 2. $A'B' = \sqrt{(8 - 4)^2 + (3 - 0)^2} = \sqrt{16 + 9}$. $A'B' = 5$, equal to AB , so the segments are congruent.

Try it: $C(0, 1)$ and $D(6, 9)$ slide by $(x, y) \rightarrow (x + 3, y - 3)$. Find $C'D'$.

check: 10

2. Reflection: mirror across a line

Rules worth knowing: across the x -axis, $(x, y) \rightarrow (x, -y)$; across the y -axis, $(x, y) \rightarrow (-x, y)$; across the line $y = x$, $(x, y) \rightarrow (y, x)$. Evidence: lengths are preserved, but the figure is turned over — across $y = x$ every slope becomes its reciprocal, which is the easiest fingerprint to check.

Example: $P(1, 1)$ and $Q(7, 9)$ are reflected across the y -axis to $P'(-1, 1)$ and $Q'(-7, 9)$. Find $P'Q'$.

Step 1. To show a reflection is rigid I need a length on each side of the mirror, so compute the image length and compare it with $PQ = 10$.

Step 2. $P'Q' = \sqrt{(-7 - (-1))^2 + (9 - 1)^2} = \sqrt{36 + 64}$. $P'Q' = 10$, so $\overline{PQ} \cong \overline{P'Q'}$.

Try it: $R(1, 2)$ and $S(4, 6)$ are reflected across the x -axis. Find $R'S'$.

check: 5

3. Rotation: turn about a centre

About the origin: a quarter turn counterclockwise is $(x, y) \rightarrow (-y, x)$; a half turn is $(x, y) \rightarrow (-x, -y)$; a quarter turn clockwise is $(x, y) \rightarrow (y, -x)$. Evidence: lengths are preserved. Slopes change for every rotation *except* the half turn, where rise and run both change sign and the quotient survives.

Example: $E(1, 2)$ and $F(4, 6)$ are turned a quarter turn counterclockwise about the origin to $E'(-2, 1)$ and $F'(-6, 4)$. Find $E'F'$.

Step 1. A rotation should leave the length alone, so the check is to recompute the distance after the turn and compare it with $EF = 5$.

Step 2. $E'F' = \sqrt{(-6 - (-2))^2 + (4 - 1)^2} = \sqrt{16 + 9}$. $E'F' = 5$, unchanged, so the rotation is a rigid motion.

Try it: $G(0, 0)$ and $H(6, 8)$ are turned a quarter turn counterclockwise about the origin. Find $G'H'$.

check: 01

4. Dilation: the one that changes size

About the origin: $(x, y) \rightarrow (kx, ky)$. Lengths are multiplied by k , so a dilation is *not* rigid and the figures are similar, not congruent. Evidence: compute the ratio image length over pre-image length for **several** pairs of corresponding sides — similarity needs the same k every time, and one matching ratio proves nothing.

Example: $X(0, 0)$ and $Y(4, 3)$ are dilated about the origin by $k = 3$, giving $X'(0, 0)$ and $Y'(12, 9)$. Find $X'Y'$.

Step 1. Here the length is expected to change, so compute the image length and divide by $XY = 5$ to recover the scale factor.

Step 2. $X'Y' = \sqrt{12^2 + 9^2} = \sqrt{144 + 81}$. $X'Y' = 15$, and $15 \div 5 = 3 = k$ as predicted.

Try it: $X(0, 0)$ and $Z(6, 8)$ are dilated about the origin by $k = 2$. Find $X'Z'$.

check: 02